5

Schlegel Diagrams for 4-Polytopes

Now that we understand the combinatorics of 3-polytopes (do we?), the next step is to investigate 4-polytopes. Those are harder to understand, since we (i.e., most of us) lack a genuine geometric intuition for the geometry of 4-dimensional Euclidean space. Nevertheless, there are various tools available. The most prominent one is the "Schlegel diagram" of a polytope, a polytopal complex that represents most of the geometry of a 4-polytope. We will discuss polytopal complexes in some detail (they will be needed for other purposes as well), and then we get to discuss Schlegel diagrams, and some of the traps involved.

5.1 Polyhedral Complexes

Definition 5.1. A *polyhedral complex* $\mathcal{C}$ is a finite collection of polyhedra in $\mathbb{R}^d$ such that

(i) the empty polyhedron is in $\mathcal{C}$,

(ii) if $P \in \mathcal{C}$, then all the faces of P are also in $\mathcal{C}$,

(iii) the intersection $P \cap Q$ of two polyhedra $P, Q \in \mathcal{C}$ is a face both of P and of Q.

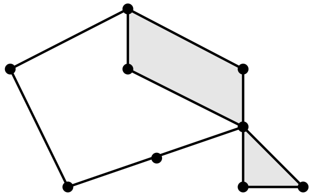
The *dimension* $\dim(\mathcal{C})$ is the largest dimension of a polyhedron in $\mathcal{C}$. The *underlying set* of $\mathcal{C}$ is the point set $|\mathcal{C}| := \bigcup_{P \in \mathcal{C}} P$.

$\mathcal{C}$ is a *polytopal complex* if all the polyhedra in $\mathcal{C}$ are bounded (polytopes).

G.M. Ziegler, *Lectures on Polytopes*, Graduate Texts in Mathematics 152,
DOI 10.1007/978-1-4613-8431-1_6,

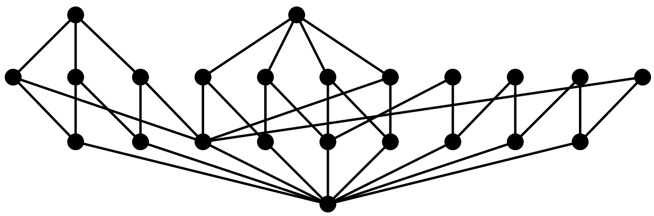
In these lectures we will almost exclusively consider polytopal complexes. The reader can work out generalizations whenever he or she feels that this is interesting.

Our sketch shows a polyhedral (in fact, polytopal) complex $\mathcal{C}_0$ of dimension 2, containing the empty polytope, 9 zero-dimensional polytopes (vertices), 11 one-dimensional polytopes (edges), and 2 two-dimensional polytopes (a triangle and a quadrangle).



Sometimes we can identify a polytopal complex $\mathcal{C}$ with its underlying set $|\mathcal{C}|$. For example, this makes sense for a polytope, since we can reconstruct the whole collection of faces from the point set P. However, often (see below) we are really interested in *subdivisions* of polytopes and polytopal complexes, and then $\mathcal{C}$ contains decisive extra information that cannot be recovered from the point set $|\mathcal{C}|$.

The combinatorial structure of a polytopal complex $\mathcal{C}$ is captured by its *face poset* $L(\mathcal{C}) := (\mathcal{C}, \subseteq)$: the finite set of polytopes in $\mathcal{C}$, ordered by inclusion. Assuming that we have a polytopal complex, we can read off the dimension function from the rank function of the face poset, by $\dim(F) = r(F){-}1$ for $F \in \mathcal{C}$. We define two polytopal complexes to be *combinatorially equivalent* if their face posets are isomorphic as posets.



Our drawing shows the poset $L(\mathcal{C}_0)$ for the complex $\mathcal{C}_0$ above. Note that $L(\mathcal{C})$ does not have a unique maximal element, unless $\mathcal{C}$ is the complex of all faces of one single convex polytope. Thus, $L(\mathcal{C})$ is not a lattice in general (although it is a finite *meet-semilattice*: it has a minimal element, and meets exist). If we adjoin an artificial maximal element $\hat{1}$, then we get a lattice $\widehat{L}(\mathcal{C}) := L(\mathcal{C}) \cup \{\hat{1}\}$.

The *maps* $f : \mathcal{C} \longrightarrow \mathcal{D}$ between polyhedral complexes are all the maps $f : |\mathcal{C}| \longrightarrow |\mathcal{D}|$ that are affine maps when restricted to polytopes in $\mathcal{C}$. Two complexes are *affinely isomorphic* if there is such a map $f : \mathcal{C} \longrightarrow \mathcal{D}$ which is a bijection between $\mathcal{C}$ and $\mathcal{D} = \{f(F) : F \in \mathcal{C}\}$. Equivalently, f has to be a bijection on the underlying sets, such that $f(F)$ is a polytope in $\mathcal{D}$ for every $F \in \mathcal{C}$. A *subcomplex* of a polytopal complex is a subset $\mathcal{C}' \subseteq \mathcal{C}$ that itself is a polytopal complex.

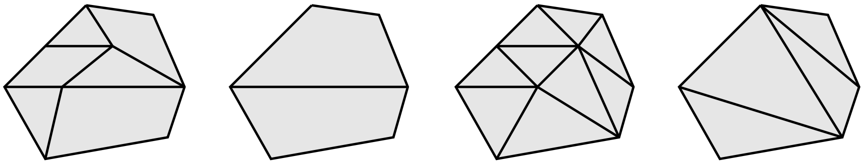
Examples 5.2. Let P be a polytope.

(i) The *complex* $\mathcal{C}(P)$ *of the polytope* P is the complex of all faces of P. The face poset of $\mathcal{C}(P)$ is the face lattice $L(P)$.

(ii) The *boundary complex* $\mathcal{C}(\partial P)$ is the subcomplex of $\mathcal{C}(P)$ formed by all proper faces of P. Thus its underlying set is $|\mathcal{C}(\partial P)| = \partial P = P \backslash \mathrm{relint}(P)$. Its face poset is $L(\partial P) := L(P) \backslash \{P\}$.

(iii) A *(polytopal) subdivision* of a polytope P is a polytopal complex $\mathcal{C}$ with the underlying space $|\mathcal{C}| = P$. The subdivision is a *triangulation* if all the polytopes in $\mathcal{C}$ are simplices. In particular, one is interested in subdivisions and triangulations without new vertices, that is, where the only zero-dimensional polytopes in the complex are the vertices of P.

Our drawing shows (from left to right, everybody smile, please!) a polytopal subdivision of a hexagon, a subdivision without new vertices, a triangulation, and a triangulation without new vertices.



The study of subdivisions of polytopes has received an enormous amount of attention in recent years, motivated by applications that range from the theory of generalized hypergeometric functions (initiated by I. M. Gel'fand; see [228]) to spline theory and to questions in computational geometry. We refer to, for example, Billera [69], Edelsbrunner [190], Pach [431], Gel'fand, Kapranov & Zelevinsky [232], Lee [354, 357], and the references in these papers.

A basic and central concept is that of "regular" subdivisions.

Definition 5.3. A subdivision $\mathcal{C}$ of a polytope $Q \subseteq \mathbb{R}^d$ is *regular* if and only if it arises from a polytope $P \subseteq \mathbb{R}^{d+1}$ in the following way.

(i) The polytope Q is the image $\pi(P) = Q$ of the polytope P, via the canonical projection map

$$\pi : \mathbb{R}^{d+1} \longrightarrow \mathbb{R}^d \qquad \begin{pmatrix} \boldsymbol{x} \\ x_{d+1} \end{pmatrix} \longmapsto \boldsymbol{x},$$

which "deletes the last coordinate."

(ii) $\mathcal{C}$ is the set of all *lower faces* of P, projected down to Q, that is,

$$\mathcal{C} \;=\; \big\{\pi(F) : F \text{ is a lower face of } P\big\},$$

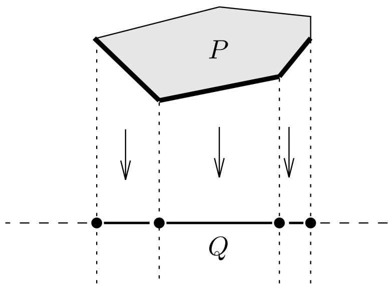
where the *lower faces* of P are the faces F that satisfy $\boldsymbol{x} - \lambda\boldsymbol{e}_{d+1} \notin P$ for each $\boldsymbol{x} \in F$ and $\lambda > 0$. Equivalently (by the Farkas lemma!), the lower faces are the faces of P of the form

$$F = \{\boldsymbol{x} \in P : \boldsymbol{c}\boldsymbol{x} = c_0\}, \quad \boldsymbol{c}\boldsymbol{x} \le c_0 \text{ valid for } P, \quad c_{d+1} < 0.$$

In other words, $\mathcal{C}$ is the family of all faces of P that can be "seen" from $-T\boldsymbol{e}_{d+1}$, for $T \longrightarrow \infty$ large enough.

If the polytope projection $\pi : P \longrightarrow Q$ is given, then we denote the subdivision $\mathcal{C}$ of Q it determines via part (ii) by $\Sigma_P(Q)$.

For $d = 1$ (subdivision of a line segment Q) this looks as follows, where the lower faces of P (4 vertices and 3 edges) are indicated by thick lines:



It suggests a reformulation: regular subdivisions arise from piecewise linear convex functions in the following way. Given the projection $\pi : P \longrightarrow Q$, the function

$$f : Q \longrightarrow \mathbb{R}, \qquad f(\boldsymbol{x}) = \min\{y \in \mathbb{R} : \binom{\boldsymbol{x}}{y} \in P\}$$

is piecewise linear and convex. (A function $f : Q \longrightarrow \mathbb{R}$ is *piecewise linear* if Q can be written as a finite union of polytopes on which f is a linear function.) For a converse, note that every piecewise linear convex function over a polytope Q determines a polytope projection, by setting

$$P := \mathrm{conv}\{\binom{\boldsymbol{x}}{f(\boldsymbol{x})} : \boldsymbol{x} \in Q\}.$$

The projection of the lower envelope of this P determines a regular subdivision $\mathcal{C}$ of Q.

Clearly, every subdivision of a line segment ($d = 1$) is regular. For $d = 2$, this is still true for subdivisions without interior points, that is, all subdivisions of a convex n-gon without new vertices are regular (Exercise 5.0).

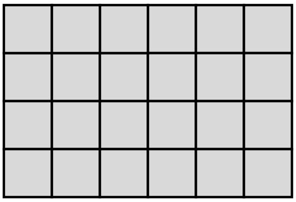
Example 5.4. For integers $z_1, \ldots, z_d \geq 1$, the *pile of cubes* $\mathcal{P}_d(z_1, \ldots, z_d)$ is the polytopal complex formed by all unit cubes with integer vertices in the *d-box*

$$B(z_1, \ldots, z_d) := \{\boldsymbol{x} \in \mathbb{R}^d : 0 \leq x_i \leq z_i \text{ for } 0 \leq i \leq d\},$$

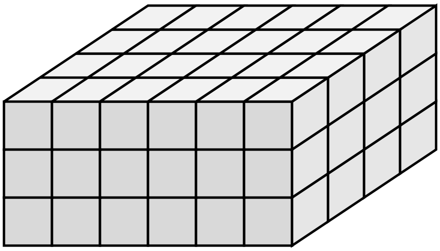
that is, the polyhedral complex formed by the set of all cubes

$$C(k_1, \ldots, k_d) = \{\boldsymbol{x} \in \mathbb{R}^d : k_i \leq x_i \leq k_i + 1\}$$

for integers $0 \leq k_i < z_i$, together with all their faces. So, our drawing represents the piles of cubes $\mathcal{P}_2(6,4)$ and $\mathcal{P}_3(6,4,3)$.



$\mathcal{P}_2(6,4)$



$\mathcal{P}_3(6,4,3)$

In particular, the pile of cubes $\mathcal{P}_d(z_1, \ldots, z_d)$ has $(z_1{+}1)\cdot \ldots \cdot(z_d{+}1)$ vertices, which are given by

$$\operatorname{vert}(\mathcal{P}_d(z_1, \ldots, z_d)) \;=\; B(z_1, \ldots, z_d) \cap \mathbb{Z}^d.$$

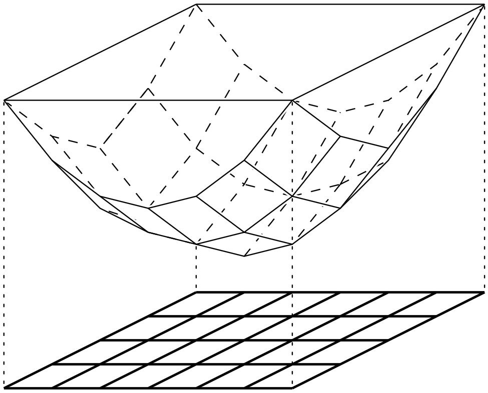
The pile of cubes is a regular subdivision of $B(z_1, \ldots, z_d)$. This is true because the pile is a "product" (Exercise 5.4) of 1-dimensional regular subdivisions. In particular, all convex functions of the form

$$f : \mathbb{R}^d \longrightarrow \mathbb{R}, \quad \boldsymbol{x} \;\longmapsto\; f(\boldsymbol{x}) = f_1(x_1) + \ldots + f_d(x_d)$$

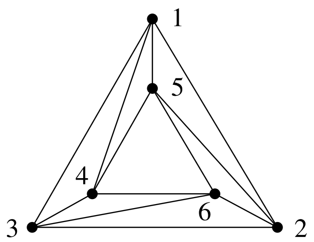
for convex functions f_i will produce suitable polytopes

$$\widetilde{\mathcal{P}}_{d+1}(z_1, \ldots, z_d) \quad := \quad \operatorname{conv}\left\{ \begin{pmatrix} \boldsymbol{v} \\ f(\boldsymbol{v}) \end{pmatrix} : \boldsymbol{v} \in B(z_1, \ldots, z_d) \cap \mathbb{Z}^d \right\}.$$

A canonical choice for f is the function $f(\boldsymbol{x}) = x_1^2 + \ldots + x_d^2 = ||\boldsymbol{x}||^2$. Our sketch tries to illustrate the construction of the 3-polytope $\widetilde{\mathcal{P}}_3(6,4)$.



To end this section, we give the following figure as a small example of a subdivision that is not regular. To see this, note that we can assume that $f(\boldsymbol{v}_4) = f(\boldsymbol{v}_5) = f(\boldsymbol{v}_6) = 0$ for the three interior vertices (by subtracting a linear function from f), and then we get a cycle $f(v_1) > f(v_2) > f(v_3) > f(v_1)$ of conditions for the three outer vertices.



Note that the subdivision $\Sigma_P(Q)$ encodes a lot of information about P, but essentially we "see only half of P" in $\Sigma_P(Q)$. Now, we'll "take a closer look" — in order to "see more."

5.2 Schlegel Diagrams

For a "close look" at polytopes, we choose a point of view $\boldsymbol{y}_F$ beyond facet F (see Section 3.1 for the definition of "beyond") and use the facet F as a "projection screen" for everything we "see" behind it.

Definition 5.5. Let P be a d-polytope in $\mathbb{R}^d$, and let $F \in L(P)$ be a facet of P, defined by the valid inequality $\boldsymbol{ax} \leq z$. We denote by

$$H := \operatorname{aff}(F) = \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{ax} = z\}$$